Business Intelligence Systems

INFO 4555 5556

Fall 2025

Retail Sales

Chapter 3



The first concept quiz was:

0

(A) A: Easier than I expected

0%

(B) B: OK

0%

(C) C: Harder than I expected

0%

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Four-Step Dimensional Design Process

- STEP 1: Select the Business Process
  - Frequently expressed as action verbs because they represent business activities
  - Typically supported by operational system
- STEP 2: Declare the Grain
  - The grain conveys the level of detail associated with the fact table measurements.
  - Answers “How do you describe a single row in the fact table?”
  - Example grains: “One row per line item on a bill from doctor”, “one row per bank account each month”

Four-Step Dimensional Design Process (cont)

- STEP 3: Identify the Dimensions
  - How do business people describe the data resulting from the business process measurement events?
  - Represent the “who, what, where, when, why and how” associated with the event
  - Common dimensions include: date, product, customer, employee, and facility
- STEP 4: Identify the Facts
  - What is the process measuring?

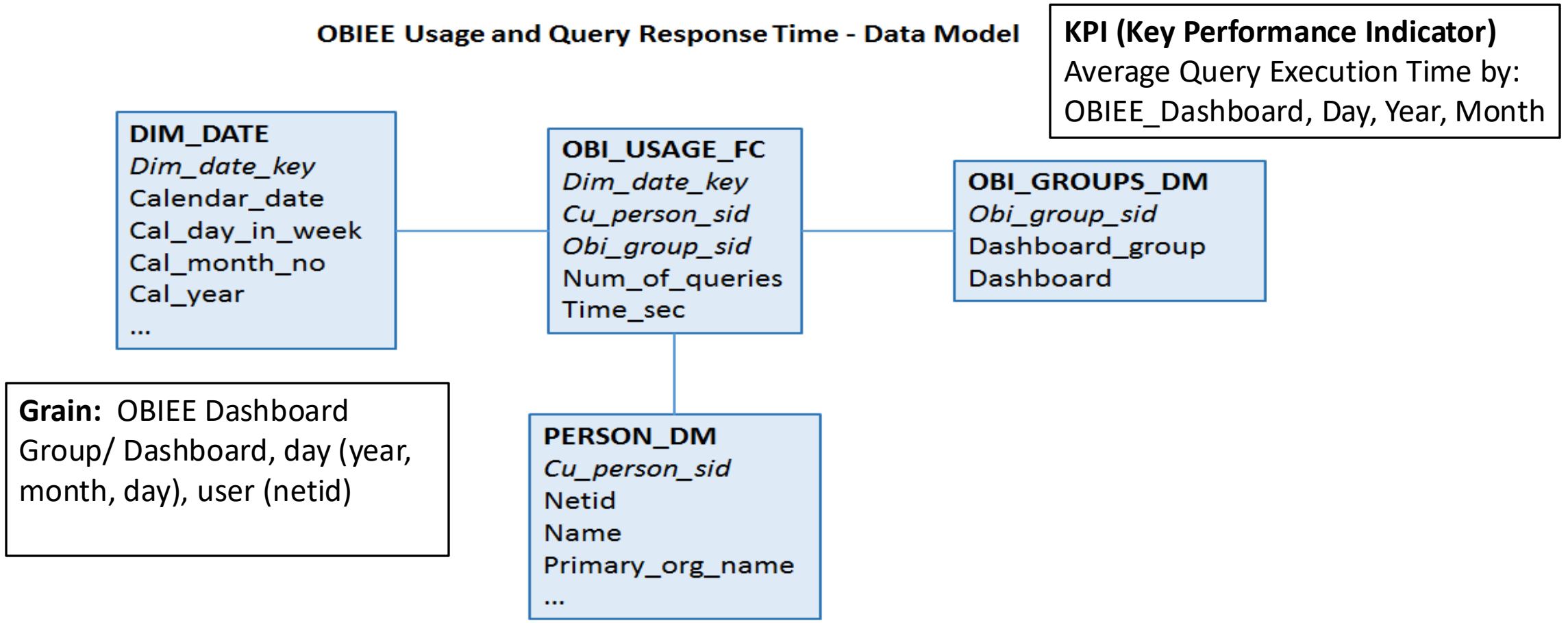
A Warning

Warning: *Do not attempt to model facts without business input. The source data realities are an important factor, but the business interviews are key to properly identifying the business process attributes. (pg 72)*

Fact Tables and Derived Fields

- Derived/calculated fields are often used in fact table to improve query performance and customer usability
- Derived/calculated fields should be part of the physical fact table if they are additive.
- Business users can define the rules for derived fields
- None additive derived fields (ex. Percentages, or ratios) should not be part of the fact table, they should be calculated by the BI tools at query time.

Example of non-additive field



Example of non-additive field (continued)

```
select d.cal_year, d.cal_month_no,og.dashboard_group, og.dashboard,
       p.primary_college_org_name,
       sum(ou.num_of_queries) number_of_queries,
       sum(ou.time_sec) total_query_time_seconds,
       (sum(ou.time_sec) / sum(ou.num_of_queries)) avg_query_wait
from obi_usage_fc ou, obi_groups_dm og, dim_date d, person_dm p
where  ou.obi_group_sid = og.obi_group_sid
       and ou.cu_person_sid = p.cu_person_sid
       and ou.dim_date_key = d.dim_date_key
group by d.cal_year, d.cal_month_no,og.dashboard_group, og.dashboard,
       p.primary_college_org_name
order by d.cal_year, d.cal_month_no,og.dashboard_group, og.dashboard,
       p.primary_college_org_name;
```

The non-additive derived attribute "avg_query_wait" should NOT be stored in Fact table, but calculated at query time to allow aggregation and drill-down.

Non-additive Measure examples

- Monthly account balance
- Ratios
- Unit price

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What is the problem with including percentages and ratios in a fact table?

A: Percentages and ratios are not measures

B: It is faster to calculate these values “on the fly"

C: Percentages and ratios are non-additive

D: Percentages and rations should be stored n dimension tables.

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Kimball Date Dimension Sample

Date Key (PK)	Date	Day of Week	Calendar Month	Calendar Quarter	Calendar Year	Fiscal Year-Month	Holiday Indicator
20170101	01/01/2017	Sunday	January	Q1	2017	F2017-01	Holiday
20170102	01/02/2017	Monday	January	Q1	2017	F2017-01	Non-Holiday
20170103	01/03/2017	Tuesday	January	Q1	2017	F2017-01	Non-Holiday
20170104	01/04/2017	Wednesday	January	Q1	2017	F2017-01	Non-Holiday
20170105	01/05/2017	Thursday	January	Q1	2017	F2017-01	Non-Holiday
20170106	01/06/2017	Friday	January	Q1	2017	F2017-01	Non-Holiday
20170107	01/07/2017	Saturday	January	Q1	2017	F2017-01	Non-Holiday

Time-of-Day Dimension

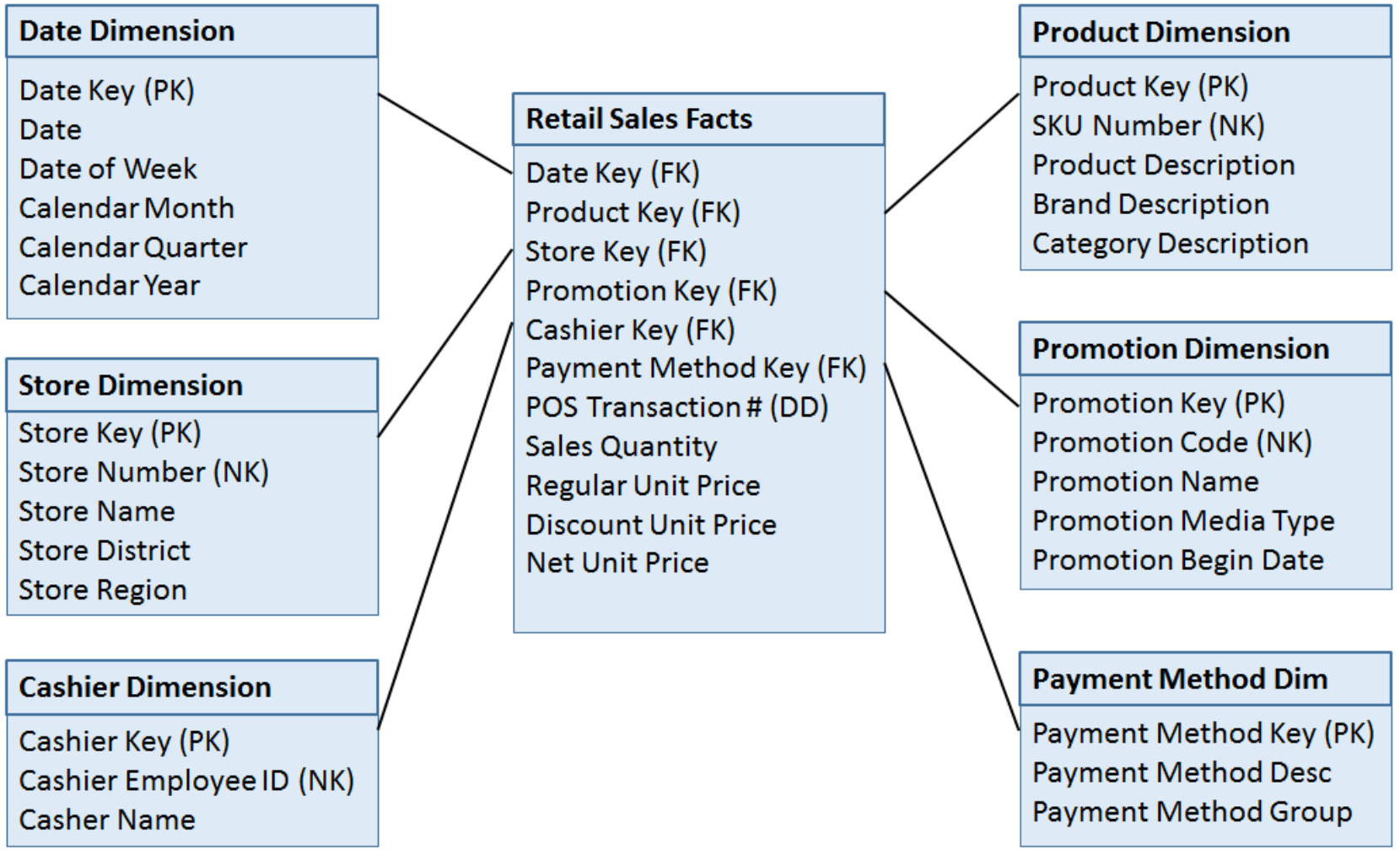
- Time of day is generally separated into its own dimension to keep the number of records minimal
- Sample attributes in a time-of-day dimension include:
  - Hour
  - Minute
  - Second (If requirements call for it)
  - Time
  - Am/Pm
  - Shift (correlate to work shifts)

Kimball Product Dimension Example

Notice the hierarchies

Product Key (PK)	Product Description	Brand Description	Subcategory Description	Category Description	Department Description
1	Baked Well Light Sourdough Fresh Bread	Baked Well	Fresh	Bread	Bakery
2	Fluffy Sliced Whole Wheat	Fluffy	Pre-Packaged	Bread	Bakery
3	Fluffy Light Sliced Whole Wheat	Fluffy	Pre-Packaged	Bread	Bakery
4	Light Mini Cinnamon Rolls	Light	Pre-Packaged	Sweeten Bread	Bakery
5	Diet Lovers Vanilla	Coldpack	Ice Cream	Frozen Desserts	Frozen Foods
6	Chocolate Lovers ½ Gal	Frigid	Ice Cream	Frozen Desserts	Frozen Foods
7	Icy Ice Cream Sandwiches	Icy	Novelties	Frozen Desserts	Frozen Foods

Kimball Retail Sales Model



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 3-12

Null values

- The operational source may contain null values for some of the dimension or fact attributes you plan to source.
- General rules for handling null values:
  - For Null values in Dimensions, add a value of “N/A” or other descriptor to indicate there is no associated value in the transactional system.
  - For Null values in Facts, leave the value as a null, to avoid miscalculation of measures. Null does not equal 0.
- Include a dimension record for “no match found” for the event where a fact record does not have an associated dimension record. Zero SID

Natural Keys and Surrogate Keys

- Natural Keys (NK) are carried over from the operational system.
- If the Natural Key contains meaning, it should be kept in its entirety as well as broken down into distinct attributes within the dimension
  - Example of SKU as a NK containing meaning (pg 84)
- Surrogate keys have no business meaning, they are sequentially assigned integers used as the primary keys for the dimension tables
- Always use a Surrogate key for the primary key, NOT the Natural Key.
  - Natural Keys are NOT necessarily unique over time

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Why is the use of surrogate keys recommended for dimension table primary keys?

A: Allow the integration of multiple source systems

B: Integer surrogate keys perform well for table joins

C: They buffer the data warehouse from operational change

D: All of the above

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Fact Table Grain – change in business rule

- Great example of how the grain of a fact table can change (or break) due to a change in business rules
- Sample of allowing multiple payment types for a single transaction would break sales model (pg 93)
  - Allowing multiple payment methods per transaction would break the declared grain of the POS model
  - Suggested creating a payment_method fact at the grain of transaction and payment method

Factless Fact Tables

- A fact table which contains NO measures only the relationship between dimensional attributes
- Good example of using a Factless Fact Table to identify items on promotion that did not sell. (pg 97)

Degenerate Dimensions

- A dimension with no supporting attributes (no associated dimensional table)
- Generally transaction control numbers from operational system
- Good to include these Operational transaction numbers in the fact table as a reference back to the source system
- Examples include: order numbers, invoice numbers, bill-of-lading numbers

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A degenerate dimension

A: Is never recommended in Dimensional Modeling

B: Is a dimension containing “problem records"

C: Is a dimension with no content except its primary key

D: Is generally the largest dimension in a dimensional model

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Durable Supernatural Keys

- A key assigned by the Data Warehouse to uniquely identify a natural key and guaranteed to persist for the life of the DW system
- Would be an additional attribute in the dimension table, does not replace the surrogate primary key.



Outriggers

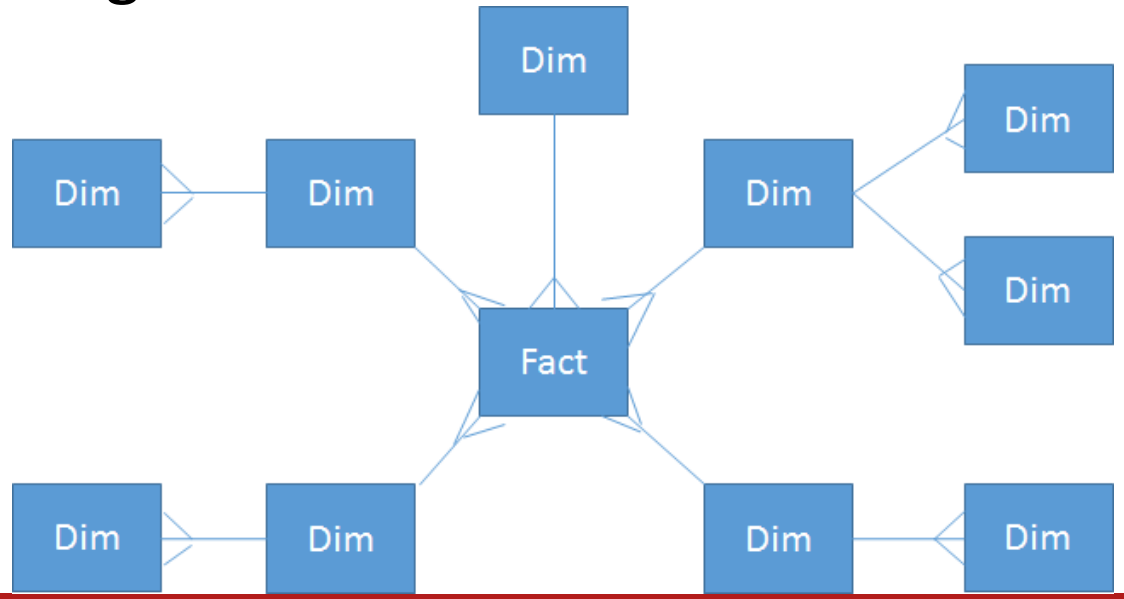
- A dimension table which joins to another dimension table.
- Must be a 1 to 1 relationship to the dimension it joins to, so the fact grain is preserved
- Example of joining a date dimension to a product dimension to add dimensionality to the “product introduction date”
- Outriggers should be used sparingly



Snowflake Schemas

- A result of normalizing a dimension table
- Tend to cause issue with the fact table grain
- NOT recommended

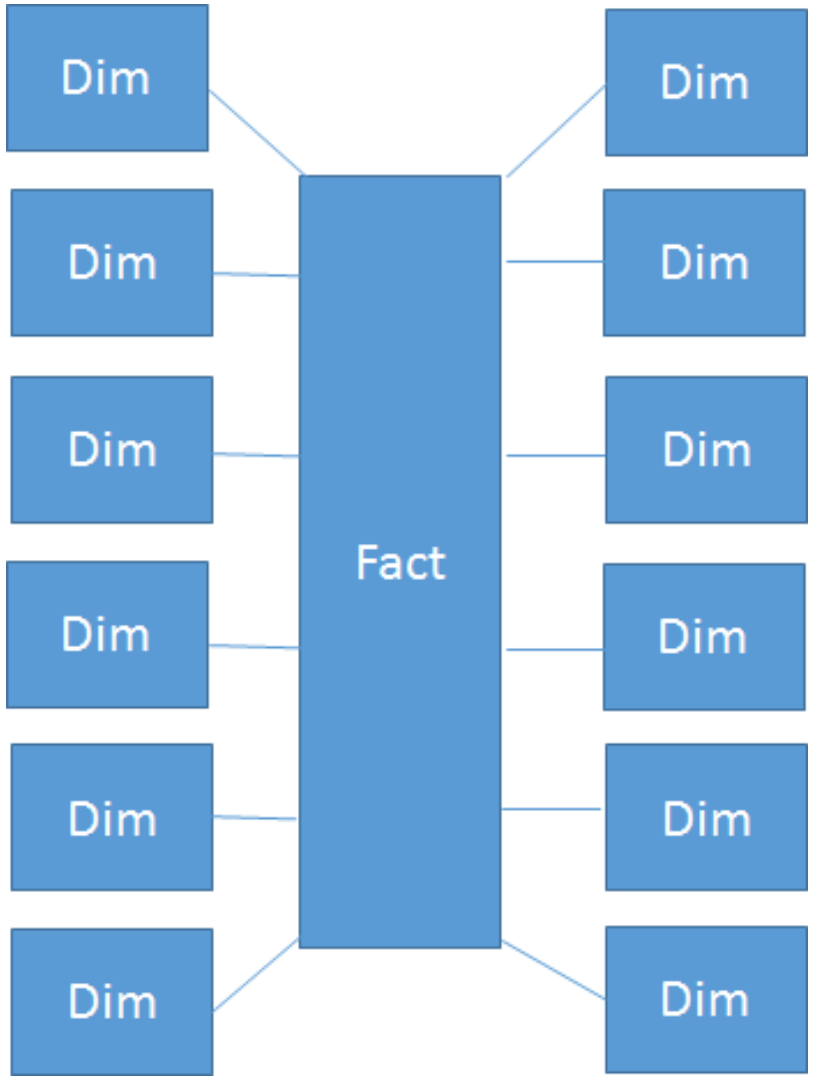




Centipede Fact Tables

- Fact Tables with too many dimensions
- Result of de-normalizing the fact table
- Don’t do this!





How are you feeling about the content of chapter 3?

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A: Very Good - I feel very comfortable with the material

0%

B: OK - Some time spent studying the slides and reading and I'm good.

0%

C: NOT Good - Dimensional Modeling is strange!

0%

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